

do-while Iteration Statement

ELEC1006: ENGINEERING COMPUTING

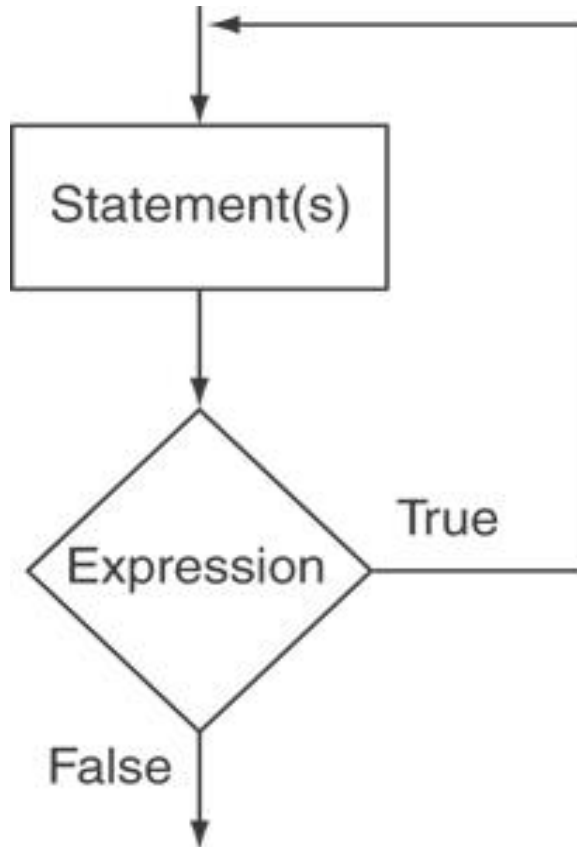
Iteration Statement (loops)

- Loops are used to repeat a statement a certain number of times when a condition is met.
- There are three types of loops:
 - *while* loop.
 - *do-while* loop. ←
 - *for* loop.

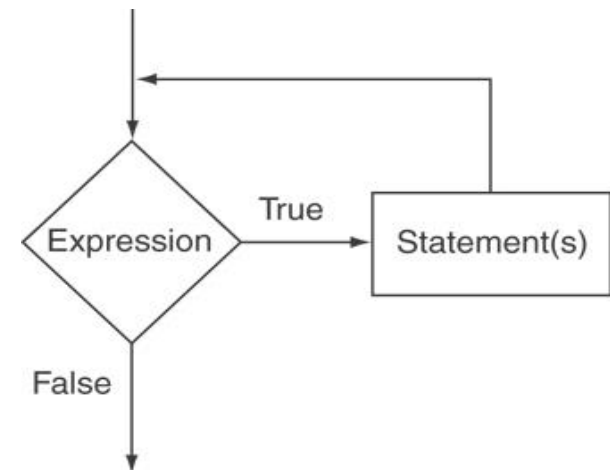
The *do-while* Loop

- The format is:
do { statements } *while* (expression or condition);
- Only difference with the *while* statement is that *do-while* evaluates the condition **after executing** the statements.

do-while Loop Logic



while Loop Logic



Example

```
// Number echo
#include <iostream>
using namespace std;
int main ()
{
    unsigned long n;
    do {
        cout << "Enter number (0 to end): ";
        cin >> n;
        cout << "You entered: " << n << "\n";
    } while (n != 0); // Must use ;

    return 0;
}
```

Notes on *do-while*

- Loop always executes at least once.
- Execution continues as long as the *expression* is true and stops the repetition when the *expression* becomes false.
- Useful in menu-driven programs to bring user back to menu to make another choice.

Menu –driven Program

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{  
    int selection;           // Selection variable
```

```
    do {
```

```
        cout << "Please make a selection: \n";           // List menu
```

```
        cout << "1) Addition\n";
```

```
        cout << "2) Subtraction\n";
```

```
        cout << "3) Multiplication\n";
```

```
        cout << "4) Division\n";
```

```
        cin >> selection;           // Get user menu selection
```

```
    } while (selection != 1 && selection != 2 && selection != 3 && selection != 4);
```

```
    // Do something with the selection here
```

```
    return 0;
```

```
}
```

More info

- [1] cplusplus.com: Statements and flow control
<https://cplusplus.com/doc/tutorial/control/>
- [2] learncpp.com: 5.6 – Do while statements
<https://www.learncpp.com/cpp-tutorial/56-do-while-statements/>